# SIMATS ENGINEERING
## ASSESSMENT TOOL 1

**COURSE CODE: CSA0411**
**COURSE NAME: Operating Systems**

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**COURSE OUTCOME COVERED**

***CO1:*** *Apply the concepts of Operating System types, structures, and system calls to demonstrate their functions in various computing environments.*

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**QUESTIONS: 5** **Total Marks:100**

Annexure A

**SHORT ANSWER TEST**

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***Code of Conduct:***

*I* K.A. Analiya (192521395) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

90/100 +

***Signature of the student:***
K. A. Analiya

Annexure A

## Short Answer Test

**Question 1 (5 Marks)**
Suppose a modern Operating System supports multiple users and processes with shared resources like CPU, memory, and I/O devices.
(a) Classify different types of Operating Systems suitable for this scenario and justify your choice.
(b) Draw the structure of an Operating System showing how resources are managed.
(c) Write a sequence of system calls required to create and execute a new process in such an environment. **(BL2)**

**Question 2 (5 Marks)**
Consider an Operating System managing multiple files and devices for concurrent users.
(a) Categorize system calls used for file and device management.
(b) Illustrate how the OS handles file operations using a layered structure diagram.
(c) Write a pseudo-code or C-based system call sequence to open, read, and close a file. **(BL3)**

**Question 3 (5 Marks)**
Assume a real-time Operating System used in an embedded system controlling industrial machinery.
(a) Identify the type of OS and explain why it is suitable for real-time applications.
(b) Draw a diagram representing the interaction between hardware, kernel, and user processes.
(c) List and explain system calls required for process synchronization in this system. **(BL2)**

**Question 4 (5 Marks)**
A distributed computing system consists of multiple interconnected computers sharing resources over a network.
(a) Classify the type of Operating System used and compare it with centralized OS.
(b) Draw the architecture of a distributed OS showing communication between nodes.
(c) Write the sequence of system calls involved in inter-process communication (IPC). **(BL3)**

**Question 5 (5 Marks)**
Consider a scenario where an Operating System must manage process scheduling, memory allocation, and device communication efficiently.
(a) Explain the major functions of an OS in handling these tasks.
(b) Draw a block diagram representing OS components and their interactions.
(c) Write system call examples for memory allocation and device management. **(BL3)**

## RUBRICS FOR CALCULATION

| Criteria | Weightage (Marks) | Level 4 – Exemplary (Full Marks) | Level 3 – Proficient | Level 2 – Developing | Level 1 – Beginning |
|---|---|---|---|---|---|
| Conceptual Understanding | 35 Marks | Demonstrates clear, complete, and in-depth understanding of OS concepts (types, structure, system calls) | Good understanding with minor gaps | Basic understanding with some misconceptions | Poor or incorrect understanding |
| Accuracy of Answers | 25 Marks | All answers are correct, precise, and relevant | Mostly correct with minor errors | Partially correct with noticeable errors | Mostly incorrect or irrelevant |
| Application of Knowledge | 15 Marks | Effectively applies concepts to scenarios/questions | Applies with minor mistakes | Limited or partial application | No meaningful application |
| Completeness of Response | 15 Marks | All parts of each question are fully answered | Most parts covered with small omissions | Some parts missing | Incomplete answers |
| Clarity & Presentation | 10 Marks | Clear, well-structured, and neatly presented answers | Understandable with minor issues | Somewhat unclear or poorly structured | Difficult to understand |

